

Model Derivations

Companion notes to “Trade Tariffs and Exchange Rates”

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These notes build the model once, for generic N , and then specialize: environment and definitions (Sections 1–3), the general linear system (Section 4), two countries (Section 5), three countries (Section 6), general symmetric N (Section 7), asymmetry (Section 8), and the canonical PTA model as a parameter restriction (Section 9). Every closed form is verified numerically in the replication package. Prose is limited to defining variables and linking equations.

Notation. Countries $i, j \in \{1, \dots, N\}$; A is the tariff imposer, B the target, C a bystander. e_{ij} = units of i ’s currency per unit of j ’s currency, so a *rise* in e_{ij} is a depreciation of i against j ; $e_{ii} = 1$ and $e_{ij} = e_{ik}e_{kj}$. Perturbations: $\nu_j \equiv d \log e_{Aj}$ with $\nu_A = 0$ (A ’s currency is the numeraire). Tariffs $\tau_{ij} \geq 0$: ad valorem, levied by i on imports from j , revenue rebated lump-sum. CES weights enter shares linearly (share-linear convention). Recurring composites:

$$m \equiv \alpha_T(1 - \alpha_D), \quad \rho_1^* \equiv 1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D), \quad D_N \equiv N\alpha_D(\eta - 1) + (N - 2)\rho + 1. \quad (1)$$

1 Environment

1.1 Technology

Each country produces a tradable T_i (only i produces it, all countries consume it) and a non-tradable N_i , from labor alone:

$$Y_{T_i} = A_{T_i} L_{T_i}, \quad Y_{N_i} = A_{N_i} L_{N_i}, \quad L_{T_i} + L_{N_i} = L_i. \quad (2)$$

Perfect competition and the normalization $P_{T_i} = 1$ (producer price in i ’s own currency) give

$$w_i = A_{T_i}, \quad P_{N_i} = \frac{w_i}{A_{N_i}} = \frac{A_{T_i}}{A_{N_i}}. \quad (3)$$

Wages are constant in own currency; all adjustment runs through exchange rates (Section 2.5).

1.2 Preferences: two-layer nest

$$C_i = C_{T_i}^{\alpha_T} C_{N_i}^{1-\alpha_T} \quad (\text{Cobb–Douglas outer, tradable share } \alpha_T) \quad (4)$$

$$C_{T_i} = \left[\alpha_D^{1/\eta} C_{T_{ii}}^{\frac{\eta-1}{\eta}} + (1 - \alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} \quad (\text{home vs. imports, elasticity } \eta, \text{ home weight } \alpha_D) \quad (5)$$

$$M_i = \left[\sum_{j \neq i} \beta_{ij}^{1/\rho} C_{T_{ji}}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} \quad (\text{across origins, elasticity } \rho, \text{ weights } \sum_{j \neq i} \beta_{ij} = 1) \quad (6)$$

The symmetric case sets $\beta_{ij} = 1/(N - 1)$. The single-elasticity CES is the special case $\rho = \eta$; Cobb–Douglas within tradables is $\eta = 1$.

1.3 Policy and closure

The consumer price of T_j in i is $p_{ij} = (1 + \tau_{ij}) e_{ij}$ (in i 's currency), with $p_{ii} = 1$. Revenue is rebated lump-sum (Section 3.2).

Assumption 1 (Closure). *There are no internationally traded assets: each country's trade balance is zero every period, $TB_i = 0$ for all i .*

Assumption 1 is the margin that pins down exchange rates. Relaxing it (steady-state net foreign asset positions) shifts the thresholds below but is not treated in these notes.

2 Prices, incomes, and exchange-rate concepts

2.1 Price indices

CES price indices in i 's currency, from (5)–(6):

$$P_{M_i} = \left[\sum_{j \neq i} \beta_{ij} p_{ij}^{1-\rho} \right]^{\frac{1}{1-\rho}}, \quad P_{T_i} = \left[\alpha_D p_{ii}^{1-\eta} + (1 - \alpha_D) P_{M_i}^{1-\eta} \right]^{\frac{1}{1-\eta}}, \quad (7)$$

and the consumer price index from (4) (κ_i collects share constants):

$$P_i = \kappa_i P_{T_i}^{\alpha_T} P_{N_i}^{1-\alpha_T}. \quad (8)$$

2.2 Definitions: the objects to be signed

$$\omega_{ij} \equiv \frac{e_{ij}w_j}{w_i} = \frac{A_{T_j}}{A_{T_i}} e_{ij} \propto e_{ij} \quad (\text{relative wage, common currency}) \quad (9)$$

$$\text{ToT}_{ij} \equiv \frac{P_{T_i}}{e_{ij}P_{T_j}} = \frac{1}{e_{ij}} \quad (\text{bilateral terms of trade, pre-tariff prices}) \quad (10)$$

$$q_{ij} \equiv \frac{e_{ij}P_j}{P_i} \quad (\text{bilateral real exchange rate}) \quad (11)$$

$$d\nu_i^E \equiv \sum_{j \neq i} w_{ij} d \log e_{ij}, \quad w_{ij} = \frac{f_{ij} + f_{ji}}{\sum_{k \neq i} (f_{ik} + f_{ki})} \quad (\text{NEER, trade weights}) \quad (12)$$

$$dq_i^E \equiv \sum_{j \neq i} w_{ij} d \log q_{ij} \quad (\text{REER, same weights}) \quad (13)$$

where f_{ij} is the value of i 's imports from j . With one factor and constant returns, (9)–(10) identify the nominal rate, the inverse common-currency relative wage, and the inverse terms of trade as the same object up to constants. All effective rates are depreciation-positive: $d\nu_i^E > 0$ means i 's currency loses value on average. In the symmetric model $w_{ij} = 1/(N - 1)$.

2.3 Wage–exchange-rate equivalence

Prices and demands depend on w_j and e_{ij} only through the products $e_{ij}w_j$. The model therefore does not separate relative wage adjustment from nominal exchange-rate adjustment: fixing $w_i = A_{T_i}$ and letting e adjust (as here) is equivalent to fixing e and letting relative wages adjust. Every statement about e_{ij} is a statement about ω_{ij} by (9).

3 Demands and equilibrium

3.1 Expenditure shares

With income I_i , the share-linear CES demands from (4)–(6) give expenditure shares of income (rows sum to α_T over tradables):

$$s_{ii} = \alpha_T \alpha_D \left(\frac{p_{ii}}{P_{T_i}} \right)^{1-\eta}, \quad (14)$$

$$s_{ij} = \alpha_T (1 - \alpha_D) \left(\frac{P_{M_i}}{P_{T_i}} \right)^{1-\eta} \beta_{ij} \left(\frac{p_{ij}}{P_{M_i}} \right)^{1-\rho}, \quad j \neq i. \quad (15)$$

Shares are tariff-inclusive: $s_{ij}I_i$ is spending at consumer prices; $s_{ij}I_i/(1 + \tau_{ij})$ is the producer-price (border) value.

3.2 Income

A fraction $\tau_{ij}/(1 + \tau_{ij})$ of spending $s_{ij}I_i$ is tariff revenue, rebated:

$$I_i = w_i L_i + \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij} I_i \implies I_i = \frac{w_i L_i}{1 - \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij}}. \quad (16)$$

The shares depend only on prices, so (16) is a closed form.

3.3 Trade balances and equilibrium

In a common currency, at producer prices (the tariff wedge is a domestic transfer),

$$\text{TB}_i = \underbrace{\sum_{k \neq i} \frac{s_{ki} I_k}{1 + \tau_{ki}}}_{\text{exports}} - \underbrace{\sum_{j \neq i} \frac{s_{ij} I_i}{1 + \tau_{ij}}}_{\text{imports}}. \quad (17)$$

By Walras' law $\sum_i \text{TB}_i = 0$: only $N - 1$ conditions are independent, matching the $N - 1$ free exchange rates. Equilibrium: given $\{\tau_{ij}\}$, the rates $\{e_{Aj}\}_{j \neq A}$ solve $\text{TB}_i = 0$ for $i \neq A$.

4 The general linear system

4.1 Linearization at a balanced free-trade baseline

Evaluate at $\tau_{ij} = 0$ with balanced trade, and normalize baseline incomes. Baseline objects: home share within tradables $h_i = \alpha_D$ (symmetric weights case: general h_i carries through identically), within-import shares $b_{ij} = \beta_{ij}$, income shares $s_{ij} = \alpha_T(1 - \alpha_D)b_{ij} = m b_{ij}$, and flow values $f_{ij} = s_{ij}I_i$. Balance requires $\sum_k f_{ki} = \sum_j f_{ij}$.

Perturb by tariffs dt_{ij} and collect exchange-rate changes $\nu_j = d \log e_{Aj}$. Log price changes in i 's own currency ($dp_{ii} = 0$):

$$dp_{ij} = \nu_j - \nu_i + dt_{ij}. \quad (18)$$

Index changes, from (7):

$$d \log P_{M_i} = \sum_{j \neq i} b_{ij} dp_{ij}, \quad d \log P_{T_i} = (1 - \alpha_D) d \log P_{M_i}. \quad (19)$$

Share changes, from (14)–(15), using $d \log P_{M_i} - d \log P_{T_i} = \alpha_D d \log P_{M_i}$:

$$d \log s_{ii} = -(1 - \eta)(1 - \alpha_D) d \log P_{M_i}, \quad (20)$$

$$d \log s_{ij} = \underbrace{(1 - \rho)(dp_{ij} - d \log P_{M_i})}_{\text{across origins}} + \underbrace{(1 - \eta) \alpha_D d \log P_{M_i}}_{\text{home vs. imports}}. \quad (21)$$

Income changes in the common currency, from (16) (revenue is first order only for tariffing coun-

tries):

$$d \log I_i = \nu_i + \sum_{j \neq i} s_{ij} dt_{ij}. \quad (22)$$

Differentiating (17), each flow contributes value $\times$ log change:

$$d \text{TB}_i = \sum_{k \neq i} f_{ki} (d \log s_{ki} + d \log I_k - dt_{ki}) - \sum_{j \neq i} f_{ij} (d \log s_{ij} + d \log I_i - dt_{ij}). \quad (23)$$

Equations (18)–(23) are linear in (ν, dt) . Stacking $i \neq A$ and writing $d\tau$ for the policy perturbation:

$$J d\nu = -F d\tau \implies \frac{d\nu}{d\tau} = (-J)^{-1} F = \frac{\text{adj}(-J) F}{\det(-J)}, \quad J_{il} = \frac{\partial \text{TB}_i}{\partial \nu_l}, \quad F_i = \frac{\partial \text{TB}_i}{\partial \tau} \Big|_{\nu \text{ fixed}}. \quad (24)$$

J is the adjustment mechanism (how balances respond to exchange rates); F is tariff incidence (how the tariff disturbs balances at unchanged rates). The entries of both are linear in η and ρ with baseline shares as coefficients.

4.2 Properties of J : stability and the multilateral Marshall–Lerner condition

Under gross substitutability:

$$\begin{aligned} J_{ii} &< 0 && \text{(own depreciation improves own balance: bilateral ML)} \\ J_{il} &\geq 0, \quad l \neq i && \text{(another country's appreciation improves your balance)} \\ |J_{ii}| &\geq \sum_{l \neq i} J_{il} && \text{(dominant diagonal, from homogeneity and Walras' law)} \end{aligned}$$

Then J is a stable Metzler matrix, $-J$ is an M-matrix, and

$$(-J)^{-1} \geq 0 \text{ elementwise,} \quad \det(-J) > 0. \quad (25)$$

The tatonnement $\dot{\nu}_i \propto \text{TB}_i$ (a surplus appreciates the currency) is locally stable iff all eigenvalues of J have negative real parts; (25) delivers this. The scalar case of $J_{ii} < 0$ is the textbook Marshall–Lerner condition (Section 5.3).

4.3 The sign rule

Result 4.1 (Generic sign rule). *At a stable equilibrium satisfying (25),*

$$\text{sign} \left(\frac{d\nu_i}{d\tau} \right) = \text{sign} \left(\sum_j \omega_{ij} F_j \right), \quad \omega_{ij} \equiv [(-J)^{-1}]_{ij} \geq 0. \quad (26)$$

If all impact effects F_j share a sign, all responses share it. Sign ambiguity requires a mixed impact vector: it is a property of tariff incidence F , never of a pathological adjustment mechanism J .

F is linear in ρ and $\text{adj}(-J)$ has degree $N - 2$ in ρ , so each numerator in (24) is a polynomial of degree $N - 1$ in ρ : one relevant root at $N = 3$ (Section 6), possible double crossings at $N \geq 4$.

5 Two countries

5.1 The slope

At $N = 2$ there is one balance and one rate. Specializing (18)–(23) (details: Section 7 at $N = 2$; ρ is inoperative with a single origin):

$$\boxed{\left. \frac{d \log e_{AB}}{d\tau} \right|_{\tau=0} = -\frac{\rho_1^*}{D_2}, \quad D_2 = 2\alpha_D(\eta - 1) + 1.} \quad (27)$$

The numerator is the object defined in (1), with the identity

$$\rho_1^* = 1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D) = \underbrace{\alpha_D \eta}_{\text{home switching}} + \underbrace{(1 - \alpha_D)(1 - \alpha_T)}_{\text{expenditure absorption}} > 0 \quad (28)$$

for all admissible parameters.

Result 5.1 (Conventional wisdom, $N = 2$). *Given stability ($D_2 = 2\alpha_D(\eta - 1) + 1 > 0$, guaranteed for $\eta \geq 1$), a tariff appreciates the tariffing country's currency unconditionally: $\rho_1^* > 0$ by (28).*

5.2 Exact solution in the Cobb–Douglas case

At $\eta = 1$ shares are constant: $s_{AB} = s_{BA} = m$. From (16), $I_A = w_A L_A / [1 - \frac{\tau}{1+\tau} m]$, and $TB_A = 0$ in (17) reads $e m w_B L_B = m I_A / (1 + \tau)$:

$$\boxed{e(\tau) = \frac{w_A L_A}{w_B L_B} \cdot \frac{1}{1 + (1 - m)\tau}} \quad \left. \frac{d \log e}{d\tau} \right|_{\tau=0} = -(1 - m) = -\frac{\rho_1^*}{D_2} \Big|_{\eta=1}, \quad (29)$$

strictly decreasing in τ globally: the linearized result (27) is exact-in-sign at all tariff levels in the Cobb–Douglas case.

5.3 The Marshall–Lerner condition

Reduced form: export volume $X(p^*)$ with $p^* = 1/e$, import volume $M(p)$ with $p = (1 + \tau)e$, elasticities $\varepsilon_X \equiv -X'p^*/X \geq 0$, $\varepsilon_M \equiv -M'p/M \geq 0$. The balance in A 's currency at producer prices:

$$TB_A(e, \tau) = X(1/e) - e M((1 + \tau)e). \quad (30)$$

Differentiate at $\tau = 0$, balanced trade ($X = eM$):

$$\frac{\partial TB_A}{\partial e} = \underbrace{\frac{\varepsilon_X X}{e}}_{\text{export volume}} + \underbrace{\frac{\varepsilon_M M}{\text{import volume}}}_{\text{import volume}} - \underbrace{\frac{M}{\text{valuation}}}_{\text{valuation}} = M(\varepsilon_X + \varepsilon_M - 1). \quad (31)$$

A depreciation improves the balance iff $\varepsilon_X + \varepsilon_M > 1$: the volume responses must offset the one-for-one valuation loss. The tariff response at fixed e :

$$\frac{\partial \text{TB}_A}{\partial \tau} = \frac{\varepsilon_M}{1 + \tau} eM > 0 \quad \text{whenever } \varepsilon_M > 0. \quad (32)$$

Equilibrium $\text{TB}_A(e(\tau), \tau) = 0$ and the implicit function theorem give

$$\frac{de}{d\tau} = - \frac{\partial \text{TB}_A / \partial \tau}{\partial \text{TB}_A / \partial e} : \quad (33)$$

the tariff creates a surplus at the initial rate (32); the rate must move in the direction that worsens the balance; under Marshall–Lerner that direction is appreciation. The condition is necessary and sufficient because $\text{sign}(de/d\tau) = -\text{sign}(\varepsilon_X + \varepsilon_M - 1)$.

In the nested model the condition holds identically. Comparing (31) with the linearized J at $N = 2$ (per unit of import value m):

$$\varepsilon_X + \varepsilon_M - 1 = D_2 = 2\alpha_D(\eta - 1) + 1, \quad \varepsilon_X = \varepsilon_M = \alpha_D(\eta - 1) + 1 : \quad (34)$$

the stability denominator D_2 is the Marshall–Lerner margin. At $\eta = 1$ (Cobb–Douglas), $\varepsilon_X = \varepsilon_M = 1$ and the margin is exactly 1.

5.4 Quantities

From (9)–(11), with $\nu_B = d \log e_{AB}$:

$$d \log \omega_{AB} = \nu_B, \quad d \log \text{ToT}_{AB} = -\nu_B, \quad d \log q_{AB} = (1 - 2m) \nu_B - m d\tau. \quad (35)$$

The real-rate expression is the $N = 2$ case of (62) below: the tariff appreciates A in real terms directly (it raises A ’s consumer prices at any e), and nominal pass-through is attenuated by the two countries’ tradable exposures. A tariff therefore appreciates the currency, improves the terms of trade, raises the relative wage, and appreciates the real exchange rate: at $N = 2$ every definition of “the exchange rate” moves together.

6 Three countries

6.1 The linear system

Set $N = 3$, symmetric weights $\beta_{ij} = 1/2$, equal sizes, and perturb by $d\tau = dt_{AB}$. Baseline shares $s_{ii} = \alpha_T \alpha_D$, $s_{ij} = m/2$. From (18)–(19):

$$d \log P_{MA} = \frac{1}{2}(\nu_B + \nu_C + d\tau), \quad d \log P_{MB} = \frac{1}{2}\nu_C - \nu_B, \quad d \log P_{MC} = \frac{1}{2}\nu_B - \nu_C, \quad (36)$$

(own-currency), and $d \log I_A = \frac{m}{2} d\tau$, $d \log I_B = \nu_B$, $d \log I_C = \nu_C$ (common currency). Substituting into (23) for $i = B, C$ and collecting terms in $(\nu_B, \nu_C, d\tau)$ yields, with $D \equiv D_3 = 3\alpha_D(\eta - 1) + \rho + 1$,

$$J = -\frac{mD}{4} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad F = -\frac{m}{4} \begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix}. \quad (37)$$

The elasticities enter J only through the scalar D ; the matrix part is pure counting. Inverting with $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$:

$$\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = (-J)^{-1} F d\tau = -\frac{1}{3D} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau = \frac{1}{3D} \begin{pmatrix} -(\rho + 3\rho_1^*) \\ \rho - 3\rho_1^* \end{pmatrix} d\tau. \quad (38)$$

Result 6.1 (Threshold). *At the symmetric point, with $D > 0$ (local stability, guaranteed for $\eta \geq 1$),*

$$\frac{d \log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_1^*}{3D} < 0, \quad \boxed{\frac{d \log e_{AC}}{d\tau} = \frac{\rho - 3\rho_1^*}{3D}} \quad \frac{d \log e_{BC}}{d\tau} = \frac{2\rho}{3D} > 0. \quad (39)$$

A appreciates against the target B for all parameters; A depreciates against the untariffed C iff $\rho > \rho_3^ \equiv 3\rho_1^*$; B depreciates against C for all parameters.*

In the sign rule (26), $(-J)^{-1} = \frac{4}{3mD} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and

$$\frac{d\nu_C}{d\tau} \propto F_B + 2F_C \propto \rho - 3\rho_1^*: \quad (40)$$

the threshold is the point where C 's own diversion gain ($F_C > 0$ iff $\rho > \rho_1^*$) outweighs its exposure to B 's loss ($F_B < 0$).

6.2 Impossibility under a single elasticity

Impose $\rho = \eta = \sigma$. Then

$$\rho_3^* - \sigma = 3\rho_1^*|_{\eta=\sigma} - \sigma = \sigma(3\alpha_D - 1) + 3(1 - \alpha_D)(1 - \alpha_T). \quad (41)$$

Result 6.2 (Impossibility). *If $\alpha_D \geq 1/3$, then $\rho_3^* > \sigma$ for every $\sigma > 0$: with one elasticity, the reversal against C is unreachable. At $\alpha_D = 1/3$, $\alpha_T = 3/4$ (the symmetric Cobb–Douglas calibration): $\rho_3^* - \sigma = \frac{1}{2}$, $D = 2\sigma$, and*

$$\frac{d \log e_{AC}}{d\tau} = \frac{\sigma - \rho_3^*}{3D} = -\frac{1}{12\sigma} \rightarrow 0^-. \quad (42)$$

6.3 Bilateral versus effective: the NEER

Decompose the solution (38) into an aggregate and a relative component:

$$\underbrace{\frac{\nu_B + \nu_C}{2}}_{\text{NEER of A}} = -\frac{\rho_1^*}{D} d\tau, \quad \underbrace{\nu_B - \nu_C}_{\text{relative}} = -\frac{2\rho}{3D} d\tau. \quad (43)$$

The cross-origin elasticity ρ cancels from the aggregate component and fully controls the relative one.

Result 6.3 (NEER preservation). *With symmetric weights, $d\nu_A^E/d\tau = -\rho_1^*/D < 0$ for all ρ : a unilateral tariff always appreciates the tariffing country's effective exchange rate, even when $\rho > \rho_3^*$ reverses the bilateral rate against C . The conventional wisdom fails bilaterally but survives in aggregate, and by (28) the aggregate sign is unconditional.*

6.4 Real exchange rates

From (11) and (8), $d\log q_{Aj} = \nu_j + \alpha_T(d\log P_{T_j}^{\text{own}} - d\log P_{T_A}^{\text{own}})$ with own-currency indices (19). Both bilateral real rates take one form (the $N = 3$ case of (62)):

$$d\log q_{Aj} = \left(1 - \frac{3m}{2}\right) \nu_j - \frac{m}{2} d\tau, \quad j = B, C. \quad (44)$$

Setting $d\log q_{AC} = 0$ with ν_C from (38): $(2 - 3m)(\rho - 3\rho_1^*) = 3mD$, linear in ρ , so

$$\boxed{\rho_q^* = \frac{(2 - 3m) 3\rho_1^* + 3m[3\alpha_D(\eta - 1) + 1]}{2(1 - 3m)}} \quad (\rho_q^* > \rho_3^* \text{ for } m \in (0, \frac{1}{3})). \quad (45)$$

At the benchmark ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, $m = 0.08$): $\rho_3^* = 3.96$, $\rho_q^* = 4.93$. Averaging (44) over j gives the REER:

$$\frac{dq_A^E}{d\tau} = -\left(1 - \frac{3m}{2}\right) \frac{\rho_1^*}{D} - \frac{m}{2} < 0 \quad \text{for all } \rho \text{ } (m < \frac{2}{3}): \quad (46)$$

real effective appreciation is unconditional and strictly stronger than nominal, because the tariff also raises A 's consumer prices directly.

6.5 Tariff configurations by superposition

Responses are linear in the tariff vector: any configuration is a sum of directed bilateral shocks, with mirror shocks obtained by relabeling (38).

$$\text{broad tariff (on } B \text{ and } C): \quad \frac{d\log e_{Aj}}{d\tau} = \nu_B + \nu_C = -\frac{2\rho_1^*}{D} < 0, \quad j = B, C, \quad (47)$$

$$\text{symmetric war (with } B): \quad \left.\frac{d\log e_{AB}}{d\tau}\right|^w = 0, \quad \left.\frac{d\log e_{AC}}{d\tau}\right|^w = \frac{\rho - \rho_1^*}{D}. \quad (48)$$

In (47), ρ drops out entirely: uniform protection is non-discriminatory, so there is no cross-origin margin and every bilateral rate (hence the NEER) appreciates. In (48), the belligerents' bilateral component cancels and the bystander margin decides everything:

$$\left.\frac{d\nu_A^E}{d\tau}\right|^w = \frac{1}{2} \frac{\rho - \rho_1^*}{D} > 0 \iff \rho > \rho_1^*. \quad (49)$$

Result 6.4 (Configuration dependence). *The unilateral tariff appreciates the NEER for all ρ ; the*

reciprocal war depreciates both belligerents' NEERs iff $\rho > \rho_1^*$. Retaliation, not protection itself, is what can weaken the tariffing country's currency in aggregate.

7 General symmetric N

7.1 The reduced system

Symmetric weights $b_{ij} = 1/(N-1)$, equal sizes, $d\tau = dt_{AB}$; the $N-2$ bystanders are identical with common ν_C . From (18)–(19):

$$d \log P_{MA} = \frac{\nu_B + d\tau + (N-2)\nu_C}{N-1}, \quad d \log P_{MB} = \frac{(N-2)\nu_C}{N-1} - \nu_B, \quad d \log P_{MC} = \frac{\nu_B - 2\nu_C}{N-1}, \quad (50)$$

and $d \log I_A = \frac{m}{N-1} d\tau$, $d \log I_B = \nu_B$, $d \log I_C = \nu_C$. Substituting into (23) for B and a representative bystander and collecting terms (each row scaled by $(N-1)/m$; verified symbolically):

$$D_N \begin{pmatrix} -(N-1) & N-2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = \begin{pmatrix} (N-2)\rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau, \quad D_N = N\alpha_D(\eta-1) + (N-2)\rho + 1. \quad (51)$$

As at $N=3$, the elasticities enter the adjustment matrix only through the scalar D_N ; the matrix is pure counting, with determinant N . Inverting:

$$\frac{d \log e_{AB}}{d\tau} = - \frac{N\rho_1^* + (N-2)\rho}{N D_N} < 0, \quad (52)$$

$$\frac{d \log e_{AC}}{d\tau} = \frac{\rho - N\rho_1^*}{N D_N}, \quad (53)$$

$$\frac{d \log e_{BC}}{d\tau} = \frac{(N-1)\rho}{N D_N} > 0. \quad (54)$$

At $N=3$ these are (39); at $N=2$, (52) is (27) (with ρ absent, $D_2 = 2\alpha_D(\eta-1) + 1$).

Result 7.1 (N -country threshold). *A depreciates against each untariffed bystander iff*

$$\boxed{\rho > \rho_N^* = N\rho_1^*} \quad (55)$$

The threshold is linear in N : diverted expenditure splits across $N-2$ bystanders, each receiving weight $1/(N-1)$, so a larger ρ is needed to reverse any one bilateral rate. The bilateral reversal is dimension-dependent.

7.2 The NEER: dimension- and ρ -invariant sign

Averaging with symmetric weights:

$$\frac{d\nu_A^E}{d\tau} = \frac{\nu_B + (N-2)\nu_C}{N-1} = \frac{-[N\rho_1^* + (N-2)\rho] + (N-2)[\rho - N\rho_1^*]}{(N-1)N D_N} = \boxed{-\frac{\rho_1^*}{D_N}} \quad (56)$$

The ρ terms cancel exactly, and N affects only the magnitude through D_N . The companion relative component is pure ρ :

$$\nu_B - \nu_C = -\frac{(N-1)\rho}{ND_N} d\tau. \quad (57)$$

Result 7.2 (Aggregation). *In the symmetric model, for every $N \geq 2$ and every ρ , a unilateral tariff appreciates the tariffing country's NEER: $d\nu_A^E/d\tau = -\rho_1^*/D_N < 0$ by (28). Cross-origin substitution reallocates the adjustment across partners (57) and can reverse individual bilateral rates (53), but never the aggregate.*

7.3 Impossibility at general N

With $\rho = \eta = \sigma$:

$$\rho_N^* - \sigma = \sigma(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T) > 0 \quad \text{whenever } \alpha_D \geq 1/N. \quad (58)$$

The home-bias bound weakens with N : any calibration with home share at least $1/N$ keeps the bilateral reversal invisible to single-elasticity models.

7.4 Configurations

By superposition, as in Section 6.5:

$$\text{broad tariff (on all } N-1 \text{ partners):} \quad \frac{d \log e_{Aj}}{d\tau} = -\frac{(N-1)\rho_1^*}{D_N} \quad \forall j = \frac{d\nu_A^E}{d\tau}, \quad (59)$$

$$\text{symmetric war with } B: \quad \left. \frac{d \log e_{AB}}{d\tau} \right|^w = 0, \quad \left. \frac{d \log e_{AC}}{d\tau} \right|^w = \frac{\rho - \rho_1^*}{D_N}, \quad (60)$$

and the war NEER, $\frac{N-2}{N-1} \frac{\rho - \rho_1^*}{D_N}$, has the sign of $\rho - \rho_1^*$.

Result 7.3 (War threshold is independent of N). *In a symmetric war, every bystander appreciates against both belligerents — and both belligerents' NEERs depreciate — iff*

$$\boxed{\rho > \rho_1^* = \frac{\rho_N^*}{N}} \quad (61)$$

for any N . With e_{AB} pinned at zero by symmetry, each bystander's equation decouples and its sign is its own impact effect, which does not dilute with N . At $\eta = 1$, $\rho_1^* = 1 - m < 1$: any $\rho \geq 1$ suffices.

7.5 Real rates at general N

Both bilateral real rates take one form (verified symbolically; contains (35) and (44)):

$$d \log q_{Aj} = \left(1 - \frac{mN}{N-1}\right) \nu_j - \frac{m}{N-1} d\tau, \quad j = B, C. \quad (62)$$

Setting $d \log q_{AC} = 0$ with (53) gives the real bilateral threshold (for $m < 1/N$):

$$\rho_q^*(N) = \frac{[(N-1) - mN] N \rho_1^* + mN [N \alpha_D (\eta - 1) + 1]}{(N-1)(1 - mN)} \quad (63)$$

($\rho_q^*(3) = 4.93$, $\rho_q^*(4) = 7.34$, $\rho_q^*(5) = 10.40$ at the benchmark), and averaging (62) gives the REER:

$$\frac{dq_A^E}{d\tau} = -\left(1 - \frac{mN}{N-1}\right) \frac{\rho_1^*}{D_N} - \frac{m}{N-1} < 0 \quad \text{for all } \rho, N \text{ } (m < \frac{N-1}{N}). \quad (64)$$

7.6 Correspondence across N

object	$N = 2$	$N = 3$	general N
stability = ML	$\varepsilon_X + \varepsilon_M - 1 = D_2 > 0$	$D_3 > 0$	J stable; $\det(-J) > 0$
sufficient primitive	$\eta \geq 1$	$\eta \geq 1$	gross substitutes; $-J$ an M-matrix
sign rule	one impact, unconditional	weights $\frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$	$\text{sign} \left(\sum_j \omega_{ij} F_j \right), \omega \geq 0$
bilateral threshold	none (no by-stander)	$3\rho_1^*$	$N\rho_1^*$ (symmetric); degree- $(N-1)$ roots in general
real bilateral threshold	none	$\rho_q^*(3)$ (45)	$\rho_q^*(N)$ (63)
NEER (unilateral)	$-\rho_1^*/D_2$	$-\rho_1^*/D_3$	$-\rho_1^*/D_N$: sign invariant
war threshold	—	ρ_1^*	ρ_1^* , independent of N

8 Asymmetry

8.1 General baselines

Away from symmetry, the system (18)–(23) holds with observed shares (b_{ij}, h_i, y_i) as coefficients; nothing in Section 4 used symmetry. For $N = 3$ at a general (tariff-ridden, asymmetric) baseline, the numerator of $d \log e_{AC}/d\tau$ is a quadratic in ρ ,

$$\mathcal{N}(\rho) = c_2 \rho^2 + c_1 \rho + c_0, \quad c_2 = b_{AB} b_{AC} b_{CA} b_{CB} (1 - h_A)(1 - h_C) y_A y_C > 0 \quad (65)$$

(paper appendix): reversal always occurs for ρ large enough whenever A and C both import from multiple origins, and the operative threshold is the relevant root of $\mathcal{N}$. The symmetric threshold $3\rho_1^*$ is the special case.

8.2 Disciplined asymmetry: the κ parameterization

To compare asymmetry across N , let the target B carry a fixed multiple κ of the symmetric bilateral share, with the weight matrix kept symmetric (hence balanced):

$$b_{AB} = b_{BA} = \frac{\kappa}{N-1}, \quad b_{AC} = b_{CA} = b_{BC} = b_{CB} = \frac{1 - \frac{\kappa}{N-1}}{N-2}, \quad b_{CC'} = \frac{1 - 2b_{CA}}{N-3}. \quad (66)$$

$\kappa = 1$ is exact symmetry; $\kappa > 1$ makes B a disproportionate partner without becoming exceptional as N grows. Solving the linear system with weights (66) and NEER weights $w_{Aj} = b_{Aj}$:

$$\frac{d\nu_A^E}{d\tau} = -\frac{\kappa}{N-2} \cdot \frac{\rho_1^*}{\rho + \alpha_D(\eta-1)} + O(N^{-2}). \quad (67)$$

(Verified numerically: the ratio of the exact solution to (67) is 0.99–1.00 at $N = 100$ for $\kappa \in [0.5, 5]$; at $\kappa = 1$, (67) agrees with $-\rho_1^*/D_N$ to leading order.) The response scales linearly in the target's importance κ and approaches zero from the appreciation side: proportionally disciplined asymmetry preserves the aggregate sign at every tested configuration, while bilateral thresholds move freely with the shares in (65).

9 Canonical PTA model: competing exporters

9.1 Setup

Countries A (importer), B and C (exporters). Two goods: x and numeraire y . Endowments: A holds only y ; B and C hold x (and possibly y). Quasilinear preferences in each country,

$$U_i = u_i(x_i) + y_i, \quad u_i' > 0, \quad u_i'' < 0. \quad (68)$$

A levies specific tariffs t_B, t_C on imports of x from B and C . Let q_j be the price received by exporter j (the world price of its good) and p the domestic price of x in A . Both origins sell in A at the same domestic price:

$$p = q_B + t_B = q_C + t_C. \quad (69)$$

9.2 Demands and market clearing

From $u_i'(x_i) = (\text{local price of } x)$:

$$M_A(p) = x_A^d(p), \quad E_j(q_j) = \omega_j - x_j^d(q_j), \quad j = B, C, \quad (70)$$

with $M_A' < 0$ and $E_j' > 0$. Market clearing for x :

$$M_A(p) = E_B(p - t_B) + E_C(p - t_C). \quad (71)$$

Equation (71) determines p given (t_B, t_C) ; then q_B, q_C follow from (69).

9.3 Comparative statics of a discriminatory tariff

Totally differentiate (71) with respect to t_B , holding t_C :

$$M'_A \frac{dp}{dt_B} = E'_B \left(\frac{dp}{dt_B} - 1 \right) + E'_C \frac{dp}{dt_B} \implies \frac{dp}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} \in (0, 1). \quad (72)$$

Therefore

$$\frac{dq_B}{dt_B} = - \frac{E'_C - M'_A}{E'_B + E'_C - M'_A} < 0, \quad \frac{dq_C}{dt_B} = \frac{dp}{dt_B} > 0. \quad (73)$$

Result 9.1 (Viner–Mundell). *A discriminatory tariff on B worsens B's terms of trade and improves C's terms of trade. Equivalently, a preferential tariff cut toward B (lower t_B) improves B's and worsens C's terms of trade.*

9.4 The PTA assumptions as restrictions on the three-country threshold

An improvement in C 's terms of trade is a rise in e_{AC} (10). The canonical model signs this unconditionally from $E'_j > 0$ and $M'_A < 0$; Section 6 requires $\rho > 3\rho_1^*$. The difference is the three pieces of the threshold (benchmark values: $\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$):

$$\rho_3^* = 3\rho_1^* = \underbrace{3}_{B-C \text{ linkage}} + \underbrace{3\alpha_D(\eta - 1)}_{\text{home switching}} - \underbrace{3\alpha_T(1 - \alpha_D)}_{\text{openness credit}} = 3.0 + 1.2 - 0.24 = 3.96. \quad (74)$$

The linkage claim collects the two forces running through B – C trade: B 's income loss cuts its purchases from C , and C substitutes toward B 's cheapened goods. Each canonical assumption removes one piece. Recomputed thresholds under each restriction (severing trade changes the baseline, so the pieces are approximately additive):

restriction	threshold ρ_3^*	piece removed
none (benchmark)	3.96	—
no B – C trade ($\beta_{BC} = \beta_{CB} = 0$)	1.05	the linkage claim
no home tradables ($\alpha_D = 0$)	1.80	the home-switching claim
no non-tradables ($\alpha_T = 1$)	3.60	credit maximized (small alone)
$\alpha_D = 0$ and $\alpha_T = 1$	none: reversal for all $\rho > 0$	
no B – C trade and $\alpha_D = 0$	none: reversal for all $\rho > 0$	

Three readings:

- The largest piece is the B – C linkage: severing it alone moves the threshold from 3.96 to 1.05.
- Non-tradables are not an independent force: with the linkage severed and $\alpha_D = 0$, no threshold exists at any α_T . The non-tradable share enters only through the credit.
- Two routes reach $\rho_3^* \leq 0$: $\{\alpha_D = 0, \alpha_T = 1\}$ or $\{\alpha_D = 0, \beta_{BC} = \beta_{CB} = 0\}$. The canonical model takes both, adds quasilinearity, and in the Viner case sets $\rho = \infty$.

Result 9.2. *The canonical PTA model is the restriction of the three-country model to a subspace of structures on which $\rho_3^* \leq 0$. The unconditional Viner–Mundell sign is the threshold condition of Section 6 evaluated where it cannot bind.*

The literature maps onto the table:

- competing exporters (Bagwell–Staiger 1999; Ornelas 2005): $\beta_{BC} = \beta_{CB} = 0$, quasilinear;
- symmetric endowment blocs (Krugman 1991; Bond–Syropoulos 1996): $\alpha_D \approx 0$, $\alpha_T = 1$, one elasticity;
- Viner: $\rho = \infty$;
- Kemp–Wan 1976: none of these restrictions, and no sign claims.

Applied models (GTAP-class CGE, with a two-level Armington nest as standard practice; quantitative trade) keep all pieces of (74) positive and produce configuration-dependent signs, but report simulations, not the condition. The threshold of Section 6 is that condition. Since the decomposition (74) is N times the per-partner object ρ_1^* , the same three pieces govern every threshold in these notes: the bilateral threshold ($N\rho_1^*$), the war threshold (ρ_1^*), and the NEER response ($-\rho_1^*/D_N$).